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danielhanchen



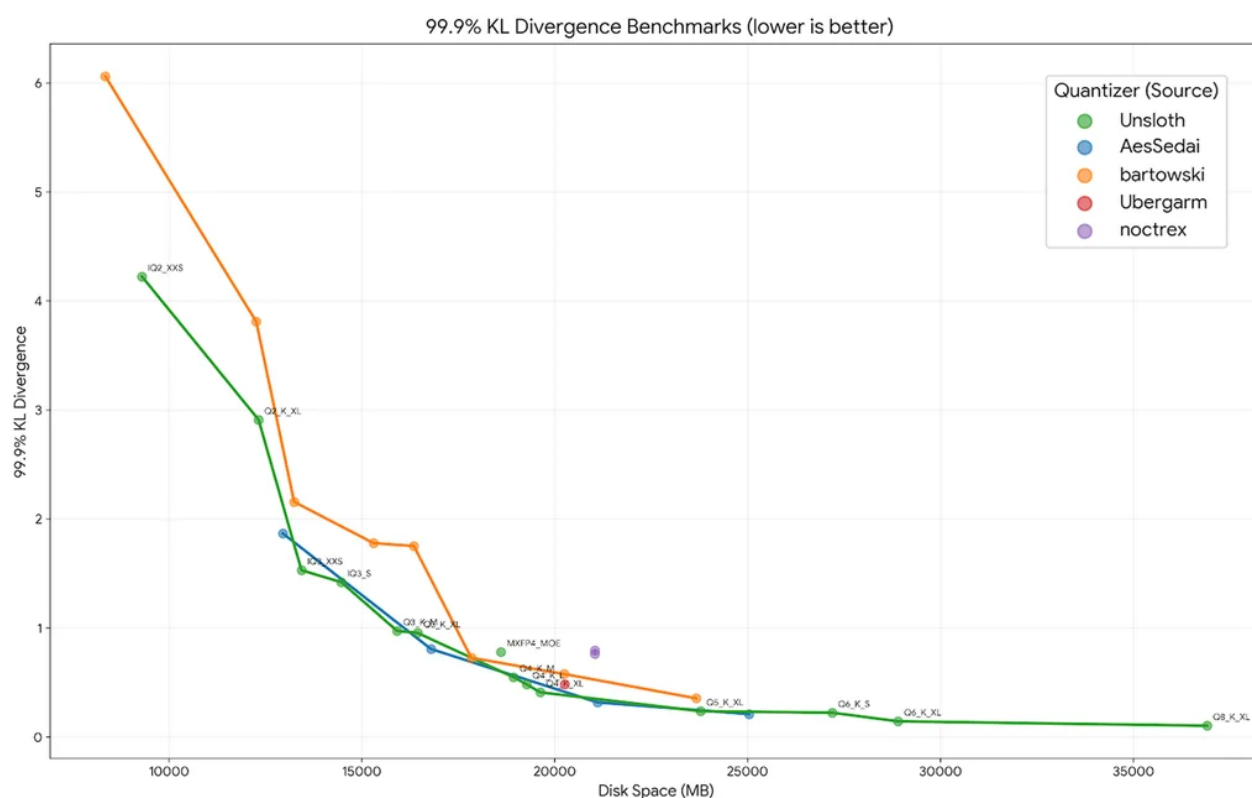
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New Qwen3.5-35B-A3B Unsloth Dynamic GGUFs + Benchmarks

Resources

Hey [r/LocalLLaMA](#)! We just updated Qwen3.5-35B Unsloth Dynamic quantizers **being SOTA** on nearly all bits. We did over 150 KL Divergence benchmarks, totally **9TB of GGUFs**. We uploaded all research artifacts. We also fixed a **tool calling** chat template **bug** (affects all quant uploaders)

- We tested Bartowski, Ubergram, AesSedai, Noctrex and our new Dynamic GGUFs
- **99.9% KL Divergence shows SOTA** on Pareto Frontier for UD-Q4_K_XL, IQ3_XXS & more.
- **Retiring MXFP4** from all GGUF quantizers: Q2_K_XL, Q3_K_XL and Q4_K_XL, except for a select few layers.
- Qwen3.5-35B-A3B GGUFs are updated to use new fixes (112B, 27B still converting, re-download once they are updated)



- Imatrix definitely helps reduce KLD & PPL.
- I quantizers (iq3_XXS, iq2_S etc) makes inference 5-10% slower.
- Quantizing ssm_out (Mamba layers) is not a good idea, and ffn_down_exps.

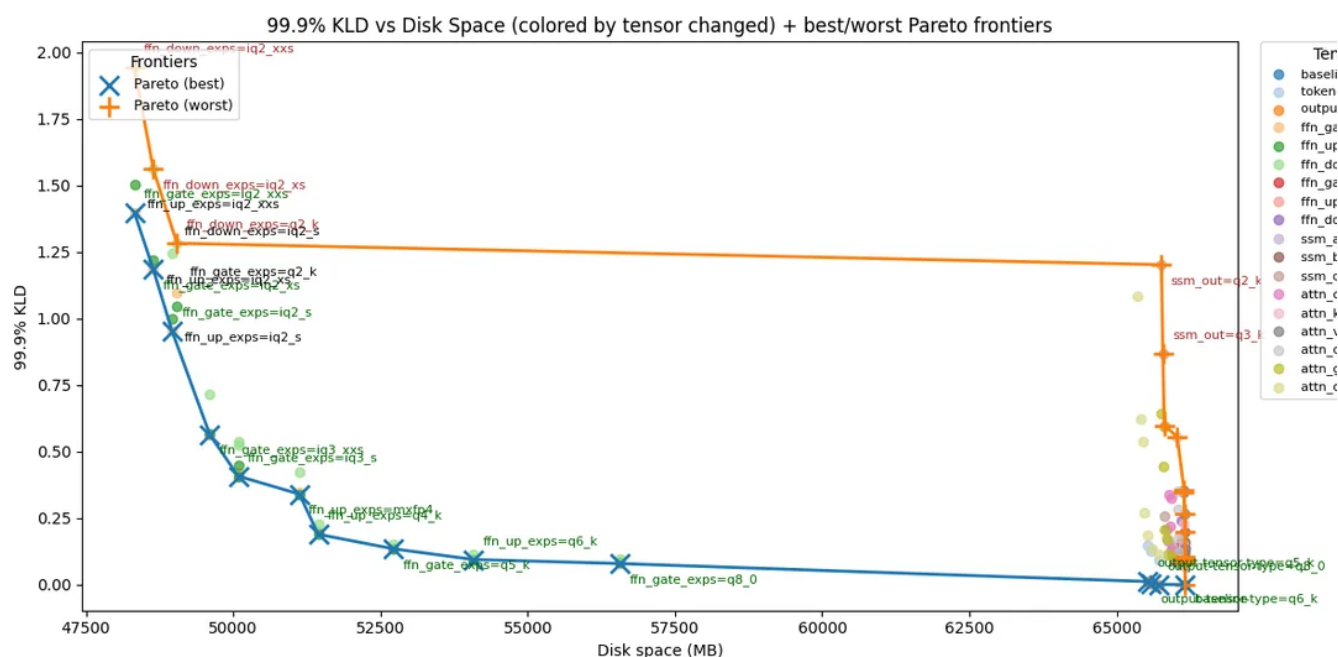
Some tensors are very sensitive to quantization

- We made over 9TB of research artifacts available for the community to investigate further on our [Experiments page](#). It includes KLD metrics and all 121 configs we tested.
- We varied bit widths across each tensor type, and generated a best and worst Pareto Frontier plot below vs 99.9% KLD.



For the worst items, ssm_out dramatically increases KLD and the disk space savings is minuscule.

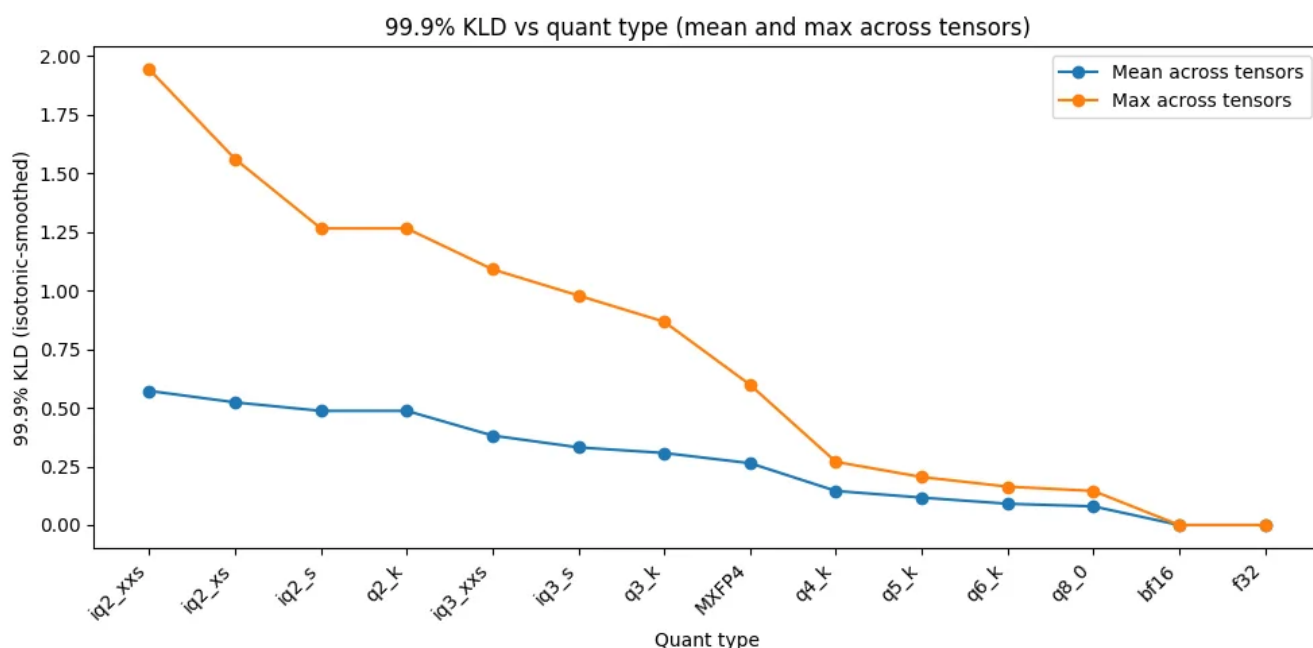
For example, ssm_out at q2_k does dramatically worse. **Quantizing any attn_* is especially sensitive** for hybrid architectures, and so leaving them in higher precision works well.



Tensor type vs bits on 99.9% KL Divergence

- We plot all quant levels vs 99.9% KLD, and sort from worst KLD to best. Quantizing ffn_* layers too heavily down is not a good idea.
- However, **some bit widths are good, especially 3bit**. - for example leaving ffn_* (down, up, gate) at around iq3_xxs seems to be best compromise on disk space and 99.9% KLD change. 2 bits cause more degradation.

MXFP4 is much worse on many tensors - attn_gate, attn_q, ssm_beta, ssm_alpha using MXFP4 is not a good idea, and rather Q4_K is better - also MXFP4 uses 4.25 bits per weight, whilst Q4_K uses 4.5 bits per weight. It's better to use Q4_K than MXFP4 when choosing between them.





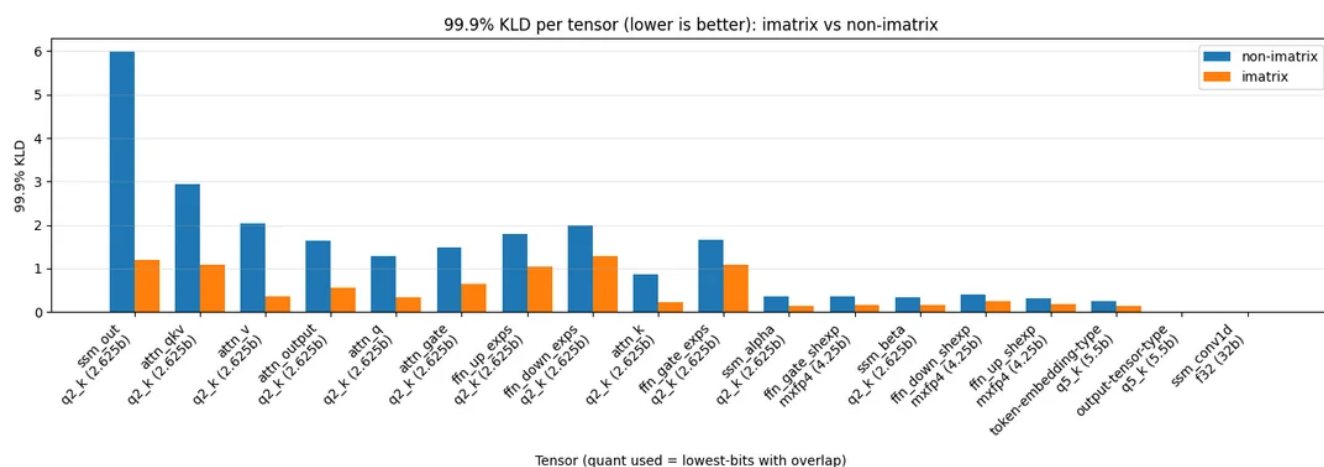
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- Imatrix definitely helps weight the quantization process in the right way. For example previously ssm_out at 2bits was really bad, however imatrix reduces the 99.9% KLD by a lot.
- Imatrix generally helps on lower bits, and works on all quants and bit widths.

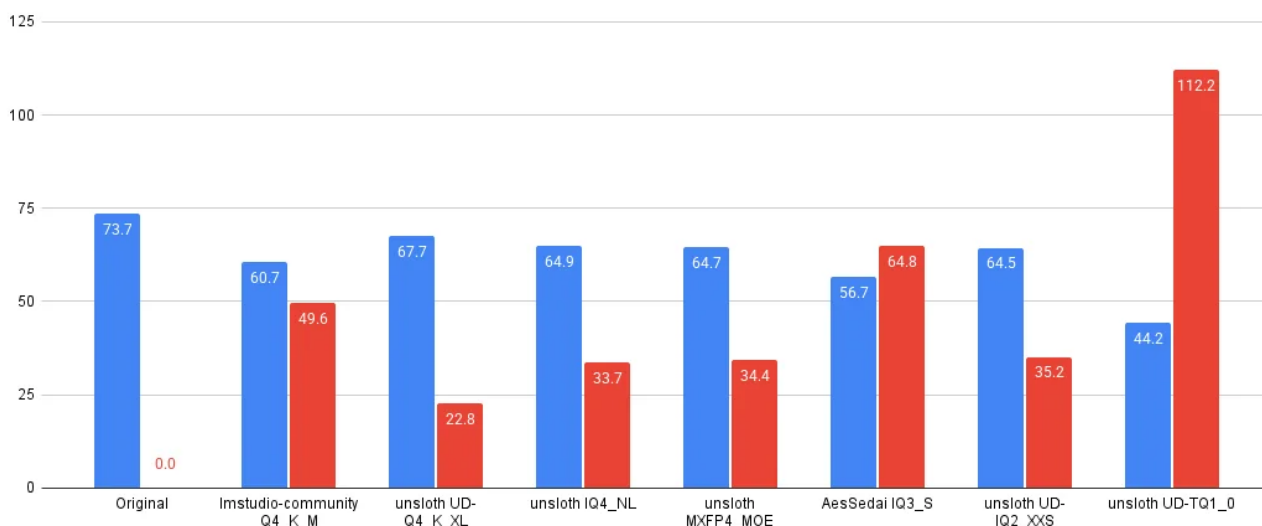


I quants (iq3_xxs, iq2_s etc) makes inference 5-10% slower, they're definitely better in terms of efficiency, but there is a tradeoff.

[Benjamin's recent MiniMax-M2.5 analysis](#) shows a case how perplexity and KLD can still be very misleading. Unsloth Dynamic IQ2_XXS **performs better** than AesSedai's IQ3_S on real world evals (LiveCodeBench v6, MMLU Pro) despite being 11GB smaller. Yet, AesSedai's perplexity and KLD benchmarks suggest the **opposite**. (PPL: 0.3552 vs 0.2441; KLD: 9.0338 vs 8.2849 - lower is better).

MiniMax-M2.5: GGUF Evaluation

Benchmark: 750 prompts from Live Code Bench (v6), MMLU Pro, GPQA, and Math500



Perplexity and KLD can also be misleading but, as precaution we replaced any MXFP4 layer. Real-world evals (LiveCodeBench v6 etc.) are much better benchmarks, but can take many days. This mismatch shows how **lower perplexity or KLD doesn't necessarily translate to better real-world performance**. The graph also shows **UD-Q4-K-XL** outperforming other **Q4** quants, while being ~8GB smaller.

This doesn't mean perplexity or KLD is useless, as they provide a *rough signal*. So, going forward, we'll publish **perplexity and KLD for every quant** so the community has some reference.



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Thank you for reading and once again for the feedback and incredible support. Huge thanks to the Qwen team as well for releasing Qwen3.5. If there's any suggestions please let us know and have a great Friday / weekend guys!

Benchmarking Details & Appreciation:

- We utilized bartowski's wonderful imatrix file to make the comparisons more fair - our Dynamic 2.0 method uses a conversational format, but we found benchmarking to be fairer if we used a more general imatrix
- We appreciated some friendly guidance from Ubergram and the community!
- For perplexity we used the below. We also use the BF16 as the base KLD file. `LLAMA_SET_ROWS=1 ./llama.cpp/llama-perplexity --flash-attn on --fit off --batch-size 16384 --ubatch-size 16384 --device {device} --model {model} --ctx-size 512`

422

158



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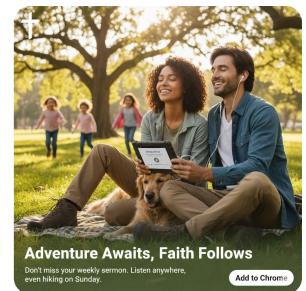


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**Round_Document6821** • 9h ago

Indeed, double checking on downstream task is a must these days since PPL and KLD is not enough.

Nice analysis from Unsloth team. Feel like this is a research itself actually :D